

Apply filters to SQL queries

Project description

The organization is working to make their system more secure. We recently found some security issues regarding the failed login attempts and employee machines. In next chapters, we will discuss the process of discovering and solving security issues in this process.

Retrieve after hours failed login attempts

We recently discovered a failed login attempt that occurred after business hours, more precisely after 18:00. In order to investigate this kind of security incident, we need to filter the log in activity in the **log_in_attempts** table. We have parameter **login_time** that shows us the time of log in attempt. Also, another important parameter we need to look at in this case is **success** (0 means fail, 1 means success). Command that will filter the failed log in attempts after business hours is:

```
SELECT * FROM log_in_attempts WHERE login_time > '18:00' AND success = 0;
```

A portion of the output shows three failed log in attempts after 18:00, one from Canada (2022-05-10), another from the USA (2022-05-11) and the last one was from Mexico (2022-05-12).

Retrieve login attempts on specific dates

Another type of security incident we had to explore was login attempts by dates, more specifically login attempts on 2022-05-08 and 2022-05-09. This was queried by next line:

```
SELECT * FROM log_in_attempts WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

For example there has been one incident since 2022-05-08 from the USA, and two log in attempts on the next day - from Canada and USA.

Retrieve login attempts outside of Mexico

Since we have determined that the suspicious activity did not originate from Mexico, we could focus on attempts from other countries by filtering the country field:

```
SELECT * FROM log_in_attempts WHERE NOT country LIKE 'Mex%';
```

We can see a few attempts from the US and Canada in the output.

Retrieve employees in Marketing

Next thing we needed to look out for were security updates on specific employee machines in the Marketing department. For this activity we needed to query the **employees** table. Our goal was to search for employees in the Marketing department that were positioned in the East building, therefore, we needed the **employees** table as mentioned previously, and two fields: **department** and **office**:

```
SELECT * FROM employees WHERE department = 'Marketing' AND office LIKE = 'East-%';
```

Some of the employees we have got here are: elarson, jdarosa and fbautist.

Retrieve employees in Finance or Sales

Similar to the previous query, we needed to perform security updates on employees in Finance or Sales department. The parameter we look out for here is the keyword **OR** - meaning we are looking either for employees in Finance or employees in the Sales department.

```
SELECT * FROM employees WHERE department = 'Finance' OR department = 'Sales';
```

The important thing to emphasize in this section is the keyword **OR** we had to use because we want to get employees from both departments, if we would try to use **AND** we would have got different results - employees that are both in Finance and Sales departments.

Retrieve all employees not in IT

In order to update systems on all employee machines (beside the machines of the IT department who already have updated their machines), we need to find all employees that are not employed in the IT department. This will be reached by using the keyword **NOT**:

```
SELECT * FROM employees WHERE NOT department 'Information Technology';
```

Summary

We have applied different filters to two different SQL tables (log_in_attempts and employees) in order to gain information regarding the security of the system. In our queries we used **AND**, **OR**, **NOT**, **LIKE** keywords, as well as wildcard sign % to filter patterns in text.